

## BPHO ROUND 0 · QUESTION PAPER

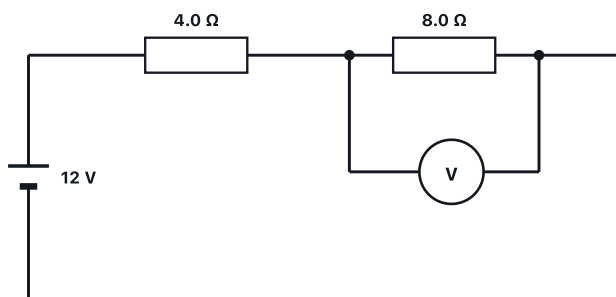
## Drill bank section 6

25 questions · 60 minutes · single answer, 5 options · **no calculator** · no negative marking*Original questions written for this course, in the Round 0 format. Not BPhO material.***Before you start**

- Answer every question. There is no negative marking, so a blank is a wasted mark.
- No calculator. Every answer here is reachable by hand, and the numbers are chosen to divide cleanly.
- Work on paper, then mark yourself against the markscheme PDF.
- The 2025 UK qualifying line was 11 out of 25, so about 44% is the target to beat.

**Q1** · A voltmeter of finite resistance loads the potential divider it is measuring, so the reading is below the unloaded value

The cell has negligible internal resistance. The voltmeter has a resistance of  $24\ \Omega$  and is connected across the  $8.0\ \Omega$  resistor, as the diagram shows.



What does the voltmeter read?

- |                |                |
|----------------|----------------|
| <b>A</b> 7.2 V | <b>B</b> 8.0 V |
| <b>C</b> 6.0 V | <b>D</b> 4.8 V |
| <b>E</b> 2.4 V |                |

**Q2** · Propagating a small fractional change through a square root, using the first-order expansion

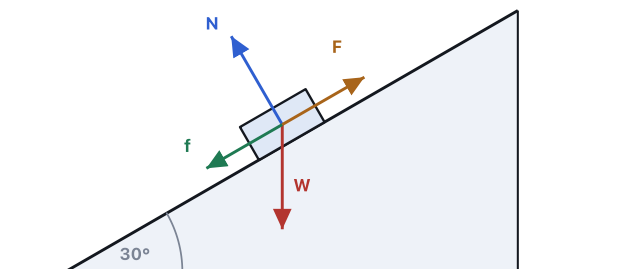
A pendulum is timed and its period is  $T$ . Its length is then increased by 4.0 %.

By approximately what percentage does its period change?

- |                 |                |
|-----------------|----------------|
| <b>A</b> 8.0 %  | <b>B</b> 4.0 % |
| <b>C</b> 2.0 %  | <b>D</b> 1.0 % |
| <b>E</b> 0.50 % |                |

**Q3** · A block pulled up a rough inclined plane at constant speed: resolving along the plane and getting the direction of friction right **C**

The block has a mass of 2.0 kg and is pulled up the slope at a constant speed by the force  $F$  shown. The coefficient of friction between the block and the slope is 0.25, and  $g$  is  $10 \text{ N kg}^{-1}$ .



What is the magnitude of  $F$ ?

- |                |               |
|----------------|---------------|
| <b>A</b> 5.7 N | <b>B</b> 14 N |
| <b>C</b> 15 N  | <b>D</b> 24 N |
| <b>E</b> 4.3 N |               |

**Q4** · A circuit with a nested parallel pair and internal resistance: working from the supply down to the current in one innermost resistor **H**

The cell has an e.m.f. of 12 V and an internal resistance of  $2.0 \Omega$ . The connecting wires have negligible resistance.

The external circuit is a  $5.0 \Omega$  resistor in series with two parallel branches. One branch is a single  $6.0 \Omega$  resistor. The other branch is a  $4.0 \Omega$  resistor in series with a parallel pair made of  $6.0 \Omega$  and  $3.0 \Omega$ .

What is the power dissipated in the  $3.0 \Omega$  resistor?

- |                  |                 |
|------------------|-----------------|
| <b>A</b> 0.24 W  | <b>B</b> 1.4 W  |
| <b>C</b> 0.080 W | <b>D</b> 0.48 W |
| <b>E</b> 0.16 W  |                 |

**Q5** · An order-of-magnitude estimate assembled from a rate, a unit conversion and a lifetime **A**

A healthy adult heart beats about 70 times a minute.

Estimate, to the nearest power of ten, how many times a heart beats in a lifetime.

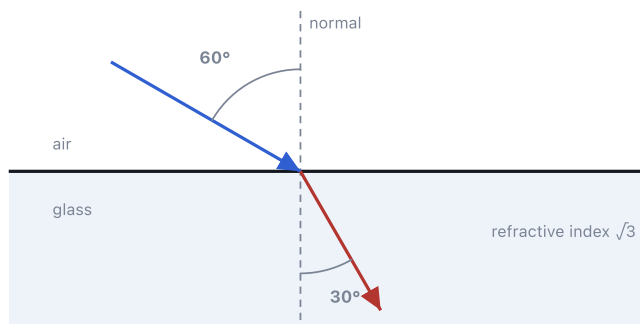
- |                    |                 |
|--------------------|-----------------|
| <b>A</b> $10^{11}$ | <b>B</b> $10^7$ |
| <b>C</b> $10^{10}$ | <b>D</b> $10^8$ |
| <b>E</b> $10^9$    |                 |





**Q10** · Refraction at a plane surface: finding the speed of light in the material from the two angles, with both measured from the normal **G**

The diagram shows a ray of light travelling in air and meeting the flat surface of a transparent material. The angle of incidence is  $60^\circ$  and the angle of refraction is  $30^\circ$ , both measured from the normal.



What is the speed of light inside the material? Take the speed of light in vacuum as  $3.0 \times 10^8 \text{ m s}^{-1}$ .

- |                                             |                                             |
|---------------------------------------------|---------------------------------------------|
| <b>A</b> $2.6 \times 10^8 \text{ m s}^{-1}$ | <b>B</b> $3.0 \times 10^8 \text{ m s}^{-1}$ |
| <b>C</b> $5.2 \times 10^8 \text{ m s}^{-1}$ | <b>D</b> $1.7 \times 10^8 \text{ m s}^{-1}$ |
| <b>E</b> $1.5 \times 10^8 \text{ m s}^{-1}$ |                                             |

**Q11** · Power dissipated inside a cell, obtained symbolically by eliminating the current between Ohm's law and the power relation **H**

A cell of e.m.f.  $\mathcal{E}$  and internal resistance  $r$  is connected to an external resistor of resistance  $R$ .

Which expression gives the rate at which energy is dissipated inside the cell?

- |                                      |                                        |
|--------------------------------------|----------------------------------------|
| <b>A</b> $\mathcal{E}^2 r / (R + r)$ | <b>B</b> $\mathcal{E}^2 r / (R + r)^2$ |
| <b>C</b> $\mathcal{E}^2 / (R + r)$   | <b>D</b> $\mathcal{E}^2 R / (R + r)^2$ |
| <b>E</b> $\mathcal{E}^2 r / R^2$     |                                        |

**Q12** · The number of nuclei decaying in an interval short compared with the half-life, using the constant-rate approximation **K**

A sample contains  $4.0 \times 10^{20}$  radioactive nuclei. The decay constant of the nuclide is  $1.0 \times 10^{-8} \text{ s}^{-1}$ .

Approximately how many nuclei decay during the first 60 s?

- |                               |                               |
|-------------------------------|-------------------------------|
| <b>A</b> $2.4 \times 10^{14}$ | <b>B</b> $4.0 \times 10^{12}$ |
| <b>C</b> $2.4 \times 10^{22}$ | <b>D</b> $2.4 \times 10^{12}$ |
| <b>E</b> $2.4 \times 10^{13}$ |                               |

**Q13** · An estimate of a daily volume: the discriminator is that every conversion from a rate per minute to a total per day must be carried out **A**

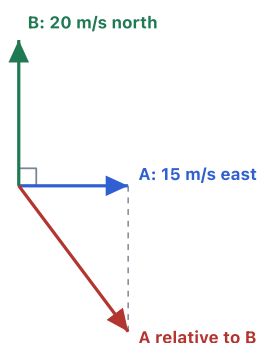
A resting adult breathes in about 0.50 litre of air with each breath, at about 12 breaths a minute.

Estimate, to the nearest power of ten, how many litres of air are breathed in during one day.

- A**  $10^1$  **B**  $10^2$   
**C**  $10^3$  **D**  $10^5$   
**E**  $10^4$
- 

**Q14** · Relative velocity of two bodies moving at right angles, obtained as a vector difference rather than a scalar one **B**

Two cyclists approach a crossroads on straight roads that meet at right angles. Cyclist P travels due east at  $15 \text{ m s}^{-1}$ . Cyclist Q travels due north at  $20 \text{ m s}^{-1}$ .



What is the speed of P relative to Q?

- A** 5.0 m/s **B** 35 m/s  
**C** 25 m/s **D** 18 m/s  
**E** 12 m/s
- 

**Q15** · A two-body system released from rest, where the driving mass reaches the floor and the sliding block is then decelerated by friction alone **C**

Block A of mass 3.0 kg rests on a rough horizontal table and is joined by a light inextensible string, passing over a frictionless pulley at the table's edge, to block B of mass 2.0 kg which hangs freely. The coefficient of friction between A and the table is 0.20, and  $g$  is  $10 \text{ N kg}^{-1}$ .

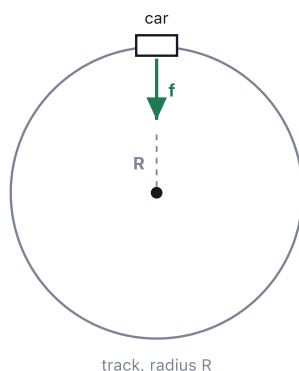
The system is released from rest. Block B falls 1.0 m to the floor, and the string then goes slack.

How much further does block A slide before coming to rest?

- A** 0.70 m **B** 2.8 m  
**C** 1.0 m **D** 1.4 m  
**E** 0.47 m
-

**Q16** · The greatest speed round a flat bend, obtained by equating the friction limit to the centripetal force and giving the answer as an expression **D**

A car of mass  $m$  goes round a flat circular bend of radius  $r$ . The coefficient of friction between the tyres and the road is  $\mu$ .



Which expression gives the greatest speed at which the car can take the bend without sliding?

- |                             |                              |
|-----------------------------|------------------------------|
| <b>A</b> $\sqrt{(\mu gr)}$  | <b>B</b> $\mu gr$            |
| <b>C</b> $\sqrt{(2\mu gr)}$ | <b>D</b> $\sqrt{(\mu gr/2)}$ |
| <b>E</b> $2\sqrt{(\mu gr)}$ |                              |

**Q17** · Linear thermal expansion of a rail, using the original length and a temperature difference in kelvin **E**

A steel rail is 2.0 m long at 20 °C. The linear expansivity of steel is  $1.2 \times 10^{-5} \text{ K}^{-1}$ .

What is the increase in the rail's length when it is heated to 120 °C?

- |                  |                 |
|------------------|-----------------|
| <b>A</b> 0.24 mm | <b>B</b> 24 mm  |
| <b>C</b> 1.2 mm  | <b>D</b> 4.8 mm |
| <b>E</b> 2.4 mm  |                 |

**Q18** · The Doppler shift for a source receding from a stationary observer, expanded to first order in the ratio of the speeds **F**

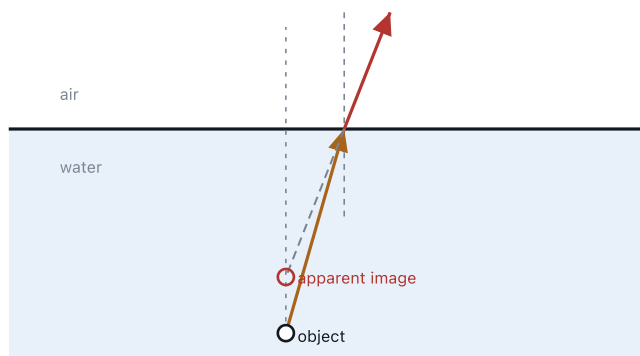
A source of sound of frequency  $f$  moves directly away from a stationary observer at a steady speed  $v$ . The speed of sound in air is  $u$ , and  $v$  is much smaller than  $u$ .

Which expression gives the frequency heard by the observer, to first order in  $v/u$ ?

- |                           |                       |
|---------------------------|-----------------------|
| <b>A</b> $f(1 + v/u)$     | <b>B</b> $f(1 - v/u)$ |
| <b>C</b> $fu/(u - v)$     | <b>D</b> $f(1 - u/v)$ |
| <b>E</b> $f(1 + v^2/u^2)$ |                       |

**Q19** · Apparent depth of an object seen through a transparent liquid, found by dividing the real depth by the refractive index G

A small stone lies on the bottom of a tank of liquid of uniform depth 3.0 m. The refractive index of the liquid is 1.5.

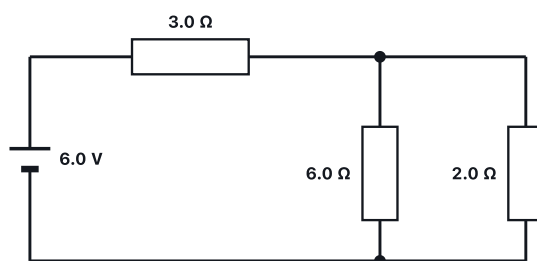


What is the apparent depth of the stone when it is viewed from directly above?

- |                |                 |
|----------------|-----------------|
| <b>A</b> 4.5 m | <b>B</b> 1.5 m  |
| <b>C</b> 2.0 m | <b>D</b> 0.67 m |
| <b>E</b> 3.0 m |                 |

**Q20** · A series resistor in front of a parallel pair: finding the current in one branch of the pair H

The cell has an e.m.f. of 6.0 V and negligible internal resistance. It is connected to a 3.0  $\Omega$  resistor in series with a parallel pair made of a 6.0  $\Omega$  and a 2.0  $\Omega$  resistor, as the diagram shows.



What is the current in the 2.0  $\Omega$  resistor?

- |                 |                |
|-----------------|----------------|
| <b>A</b> 1.3 A  | <b>B</b> 2.0 A |
| <b>C</b> 0.67 A | <b>D</b> 1.0 A |
| <b>E</b> 0.33 A |                |

**Q21** · Charge sharing between two capacitors: charge is conserved but energy is not, so a fraction of the energy is dissipated **I**

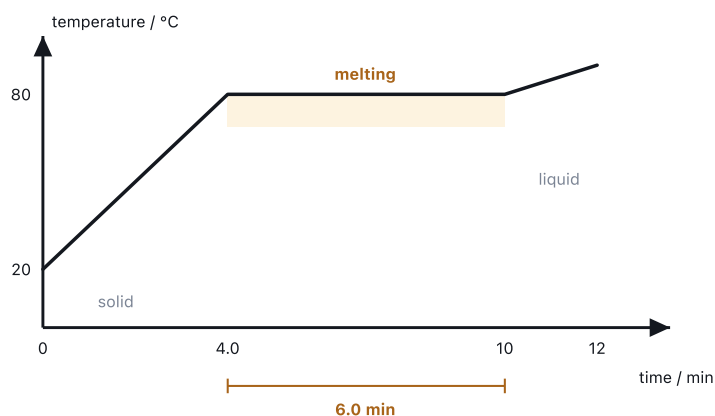
A  $4.0\ \mu\text{F}$  capacitor is charged to a potential difference of  $10\ \text{V}$  and then disconnected from the supply. It is connected directly across an uncharged  $6.0\ \mu\text{F}$  capacitor.

What fraction of the original stored energy is dissipated in the connecting wires?

- A** 60 %
- B** 40 %
- C** 50 %
- D** 0 %
- E** 20 %

**Q22** · Specific latent heat of fusion from the length of the plateau on a heating curve at constant power **J**

A  $0.30\ \text{kg}$  sample of a solid is heated at a constant rate of  $150\ \text{W}$ . Its temperature rises from  $20\ ^\circ\text{C}$  to  $80\ ^\circ\text{C}$  in  $4.0\ \text{min}$ , then stays at  $80\ ^\circ\text{C}$  for the next  $6.0\ \text{min}$  while the sample melts, and then rises again, as the graph shows.



What is the specific latent heat of fusion of the sample?

- A**  $1.2 \times 10^5\ \text{J kg}^{-1}$
- B**  $3.0 \times 10^5\ \text{J kg}^{-1}$
- C**  $1.8 \times 10^5\ \text{J kg}^{-1}$
- D**  $9.0 \times 10^4\ \text{J kg}^{-1}$
- E**  $3.6 \times 10^5\ \text{J kg}^{-1}$

**Q23** · The fraction of a radioactive sample remaining after a whole number of half-lives **K**

A radioisotope has a half-life of  $8.0\ \text{days}$ .

What fraction of the nuclei present initially remain after  $24\ \text{days}$ ?

- A**  $1/3$
- B**  $1/8$
- C**  $1/6$
- D**  $1/24$
- E**  $1/16$

**Q24** · Threshold wavelength for the photoelectric effect, obtained by equating the photon energy to the work function L

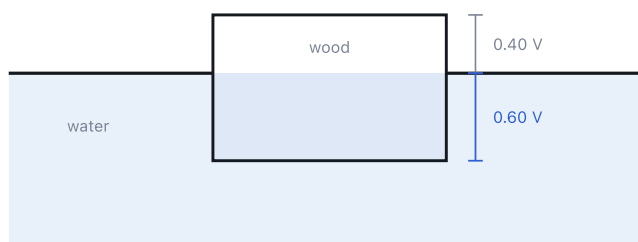
A metal surface has a work function  $\phi$ . Light of wavelength  $\lambda$  falls on it and releases electrons.

Which expression gives the longest wavelength of light that will just release an electron from this surface?

- |                                                                                                                    |                                                                          |
|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <p><b>A</b> <math>\phi/(hc)</math></p> <p><b>C</b> <math>h/(c\phi)</math></p> <p><b>E</b> <math>hc/\phi</math></p> | <p><b>B</b> <math>hc\phi</math></p> <p><b>D</b> <math>c\phi/h</math></p> |
|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|

**Q25** · A floating block: the submerged fraction is the ratio of the block's density to the liquid's, so the mass follows from the displaced volume M

A rectangular block of wood floats in water with 60 % of its volume below the surface, as the diagram shows. The density of water is  $\rho_w$  and the block's volume is  $V$ .



Which expression gives the mass of the block?

- |                                                                                                                          |                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <p><b>A</b> <math>\rho_w V</math></p> <p><b>C</b> <math>\rho_w V/0.60</math></p> <p><b>E</b> <math>\rho_w V g</math></p> | <p><b>B</b> <math>0.40 \rho_w V</math></p> <p><b>D</b> <math>0.60 \rho_w V</math></p> |
|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|